

1.1 podle def. 8.14 a pozorování 8.15 ($SL_2; (3), (2)$)

1)

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle v, v \rangle + \langle u, v \rangle + \langle v, u \rangle$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle + \langle v, v \rangle - \langle u, v \rangle - \langle v, u \rangle$$

$$\|u+v\|^2 - \|u-v\|^2 = \langle u, u \rangle - \langle u, u \rangle + \langle v, v \rangle - \langle v, v \rangle + 2\langle u, v \rangle + 2\langle v, u \rangle$$

z (SSS) $\langle v, u \rangle = \overline{\langle u, v \rangle}$ a jeliže $2\langle u, v \rangle + 2\overline{\langle u, v \rangle} = 4\operatorname{Re}(\langle u, v \rangle)$

$$4\operatorname{Re}(\langle u, v \rangle) = \|u+v\|^2 - \|u-v\|^2 \Rightarrow \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$$

2)

$$\|u-iv\|^2 = \langle u-iv, u-iv \rangle = \langle u, u \rangle + \langle iv, iv \rangle - \langle u, iv \rangle - \langle iv, u \rangle =$$

$$= \|u\|^2 - i^2\|v\|^2 - i\langle u, v \rangle + i\langle v, u \rangle$$

$$\|u+iv\|^2 = \langle u+iv, u+iv \rangle = \|u\|^2 + \|v\|^2 - i\langle v, u \rangle + i\langle u, v \rangle$$

$$\|u-iv\|^2 - \|u+iv\|^2 = -i\langle u, v \rangle + i\langle v, u \rangle + i\langle v, u \rangle - i\langle u, v \rangle + \|u\|^2 - \|u\|^2 + \|v\|^2 - \|v\|^2$$

$$= 2i\langle v, u \rangle - 2i\langle u, v \rangle = -2i\langle u, v \rangle + 2i\overline{\langle u, v \rangle}$$

protože platí: $-2i(a+bi) + 2i(a-bi) = -2ai - 2bi^2 + 2ai - 2bi^2 = -4bi^2 = 4b$

tedy $-2i\langle u, v \rangle + 2i\overline{\langle u, v \rangle} = 4\operatorname{Im}(\langle u, v \rangle)$ tedy:

$$\|u-iv\|^2 - \|u+iv\|^2 = 4\operatorname{Im}(\langle u, v \rangle)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2)$$